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HOUSE PRICE PREDICTION

The core problem statement for A.I house price predection is to accurately forecast the continuous sale price of a property, classifying it as a Regression task. The objective is to provide objective reliable property valuation to assist stakeholder like sellers in setting optimal prices and buyers in guessing market value, Predictions are driven by a mix of features, including structural details (square, footage, quality, age) location data (amenities, coordinates) and economical factors.

The Methodology begins with rigorous data preprocessing, including handling missing values, treating outliers and performing a critical log transformation on the often skewed price data, feature engineering is essential, generating new predectors like price for square foot to enhance the models understanding for Model selection, advanced esemble techniques - primarily XGBoost and Random forest are favored over single linear models. These boosting methods excel at capturing the complex non-linear

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- relationship improvement in the real estate market, Hyperparameter tuning than refines the final model configuration

Results and evaluation are measured during key regression metrix. The Root Mean Square error (RMSE) quantifies the average prediction error in dollar terms while the R^2 goals measures the proportion of price variance explained. Expected results show that advanced models consistency achieve an R^2 Score greater than 0.80 on unseen test data. Feature important analysis confirms that Square footage, Overall Quality and location are the most influential factors driving property values.